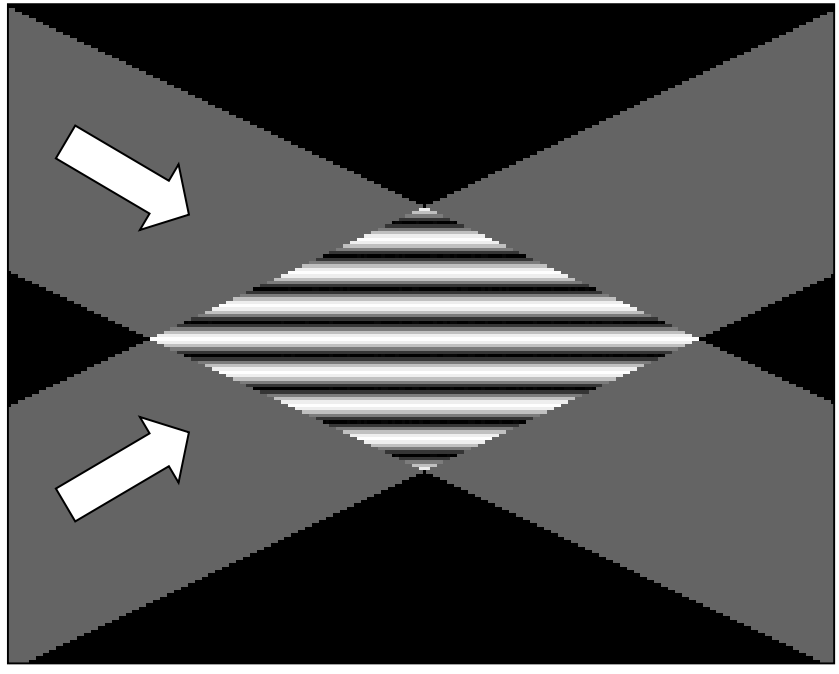
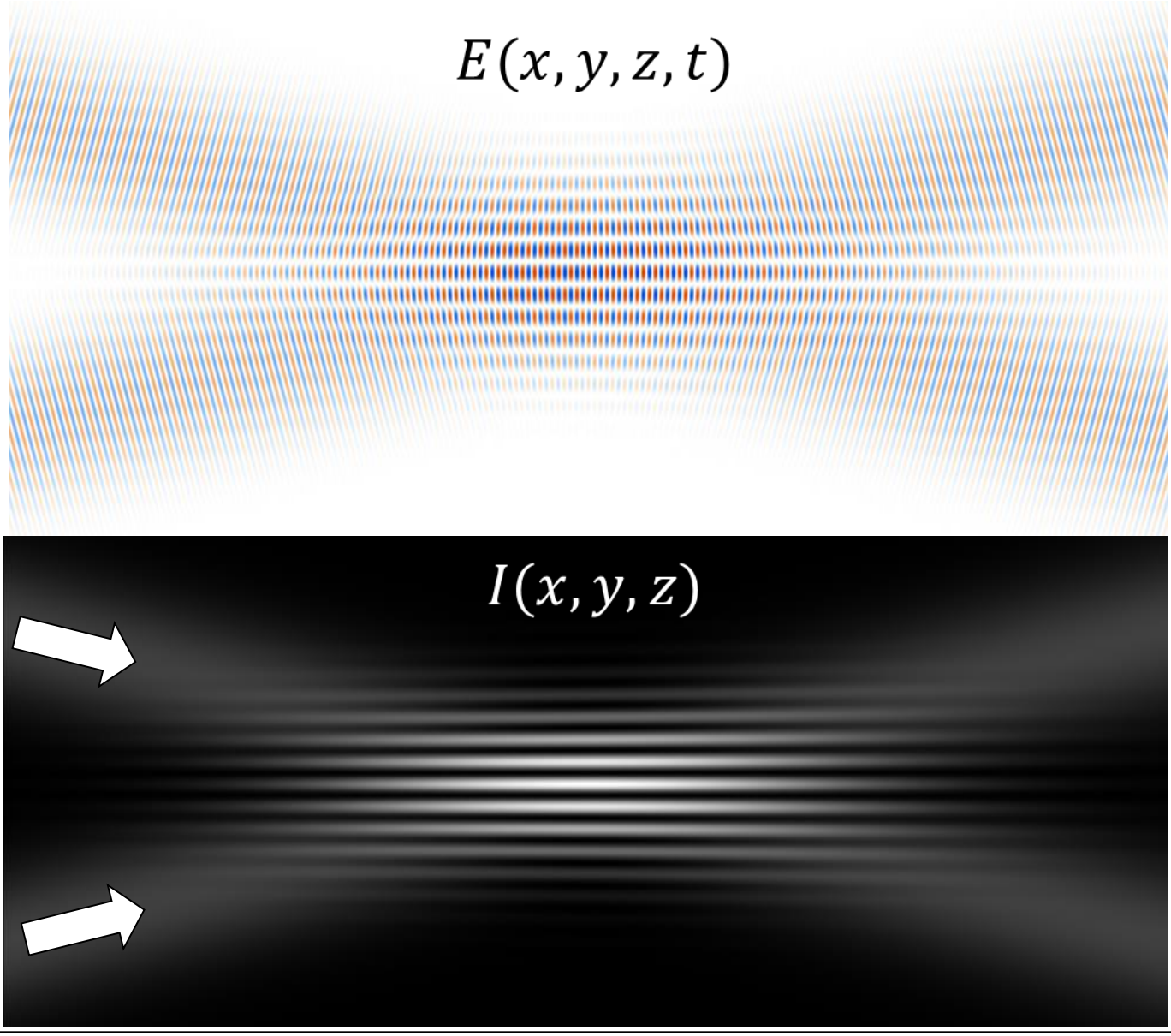
▪ Concept of interference ▪

Interference: the phenomenon when two or more discrete waves are superimposed in phase creating a spatially standing wave, which can be (generally) observed as dark-bright interference fringes in case of electromagnetic waves.

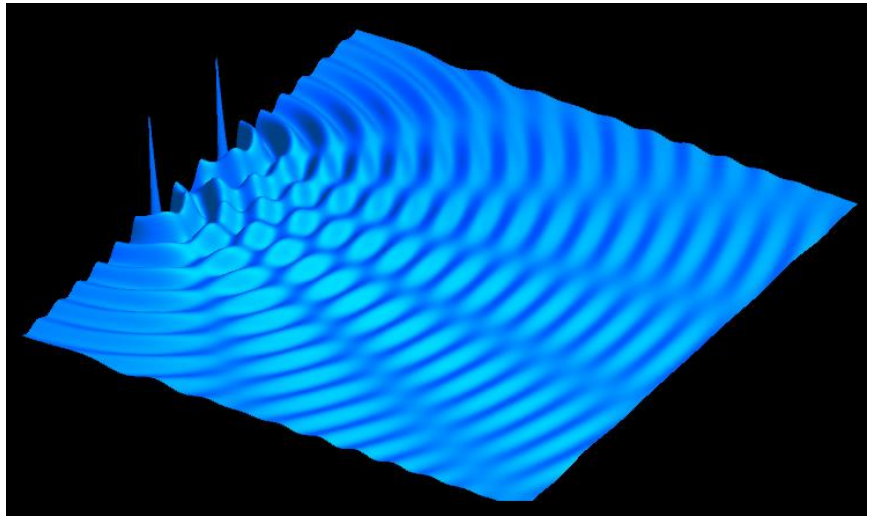


Interference of two plane waves (geometrical optical approximation)

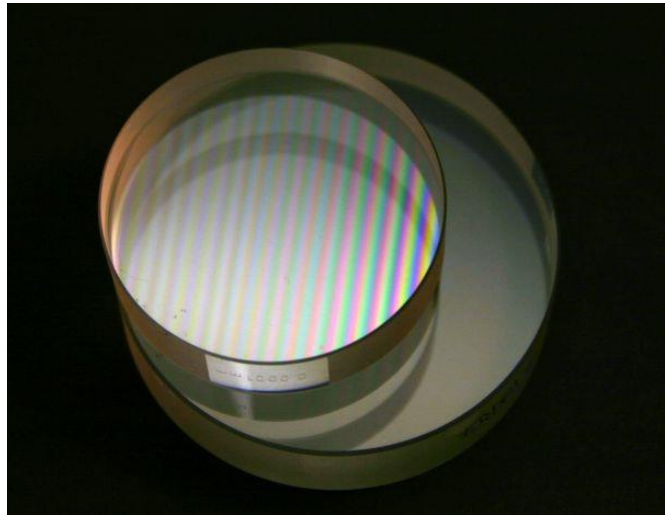
▪ Interference of Gaussian beams ▪



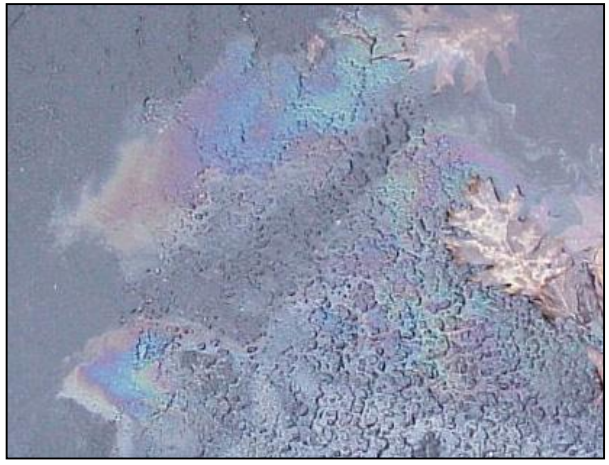
▪ Interference examples (two-beam) ▪



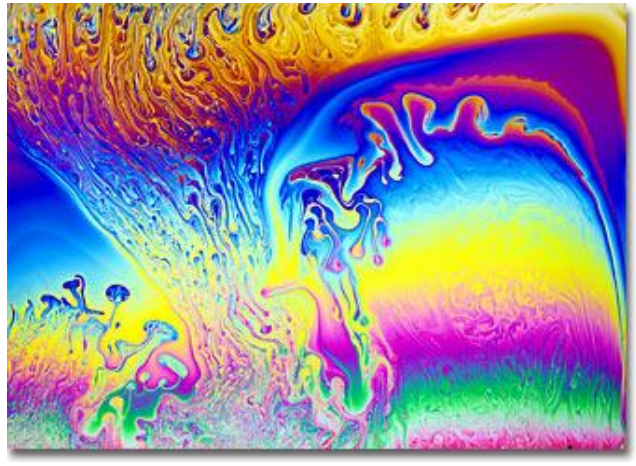
Two point sources (waves e.g. on water surface)



Newton fringes (here lines)

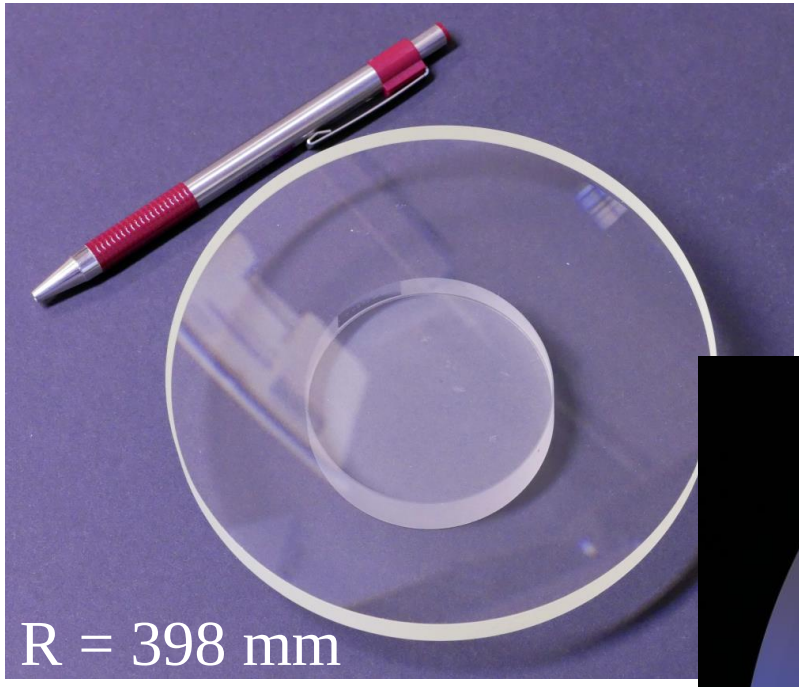


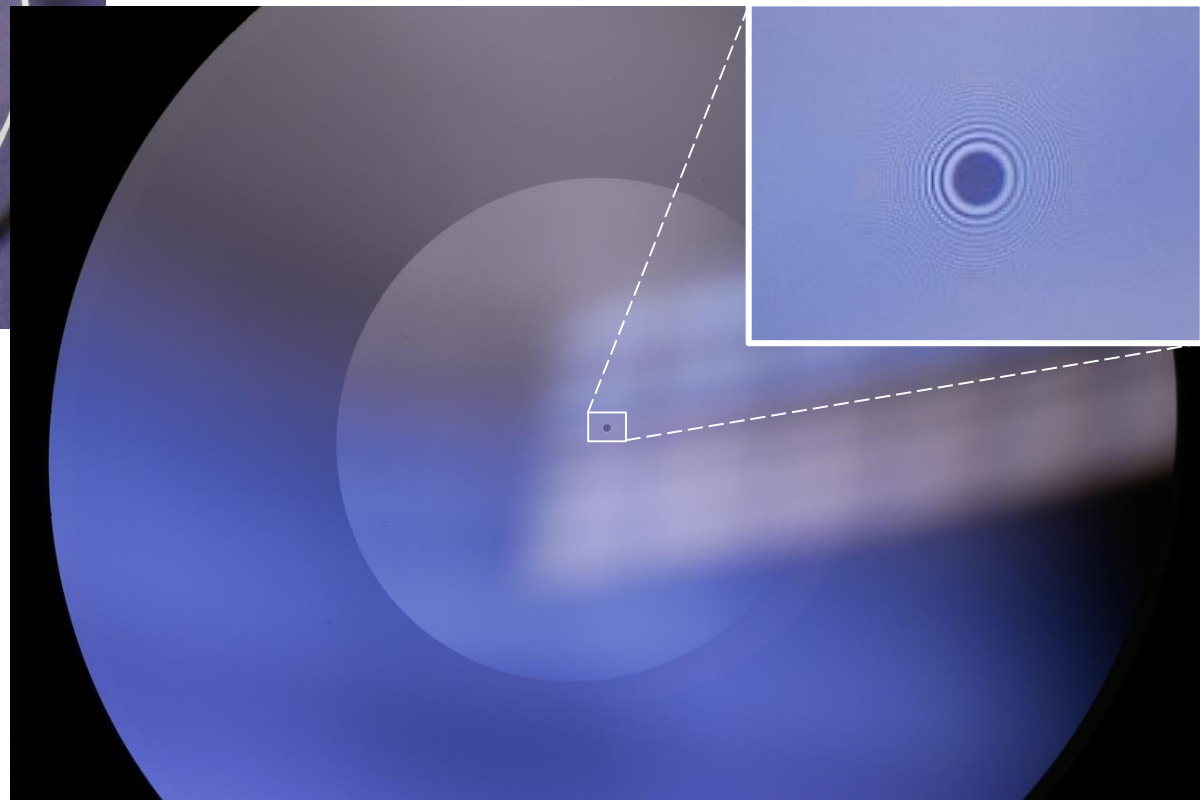
Oil film



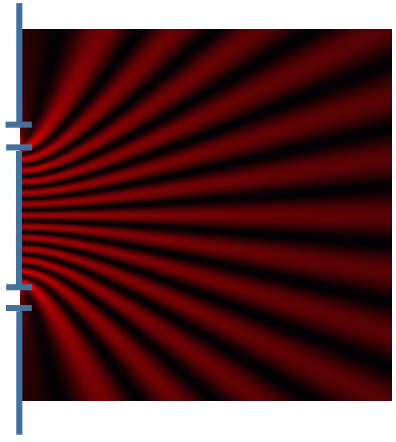
Soap bubble

▪ Newton fringes with white light ▪

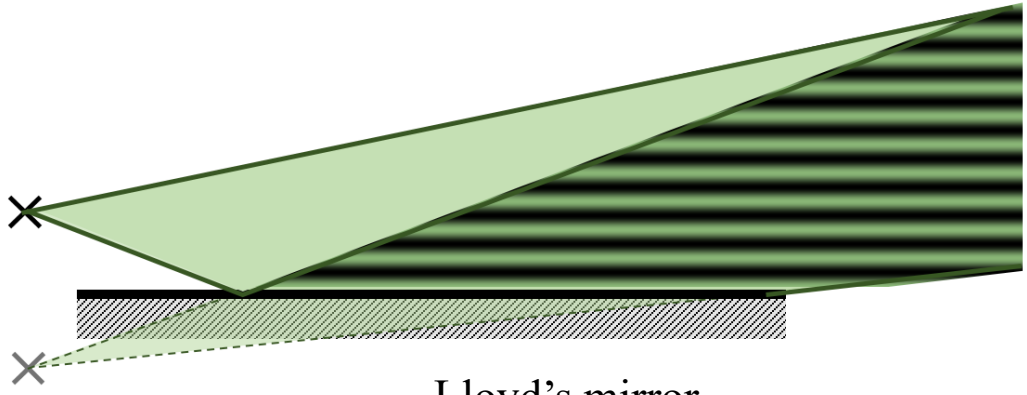




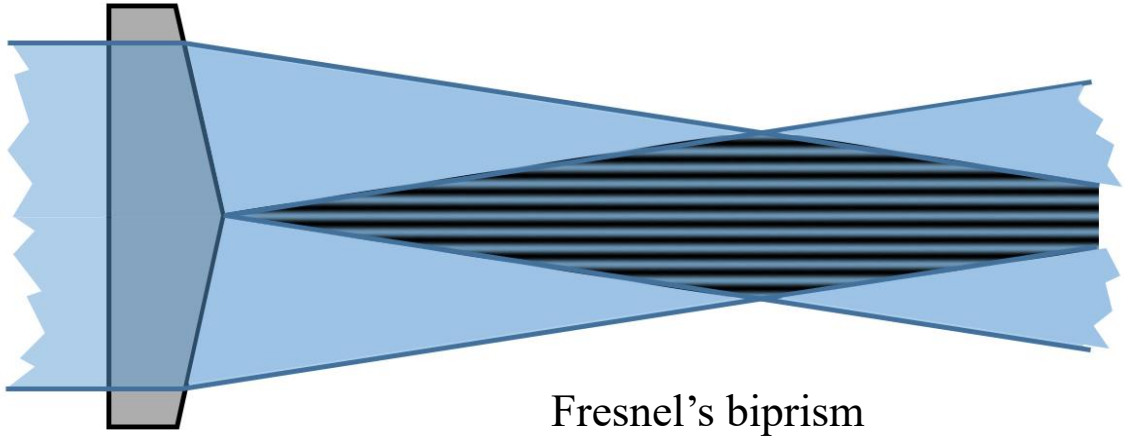
▪ Wavefront-splitting interferometers ▪



Young's two-slit interferometer

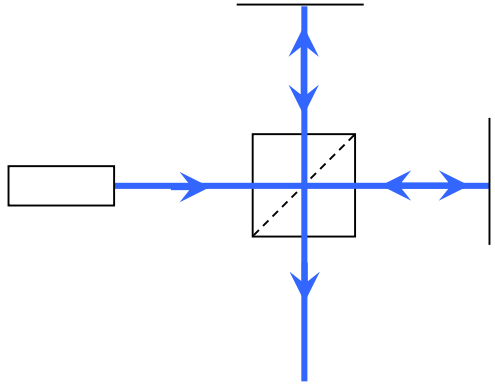


Lloyd's mirror

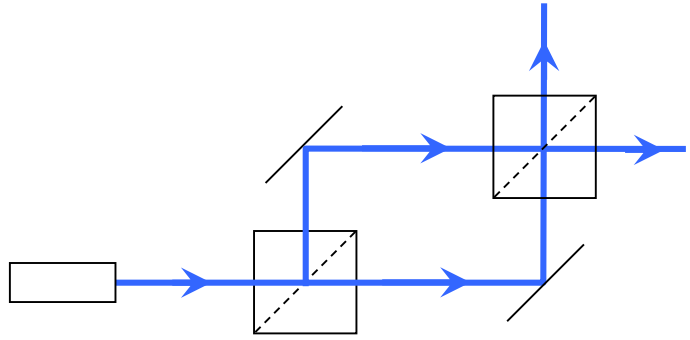


Fresnel's biprism

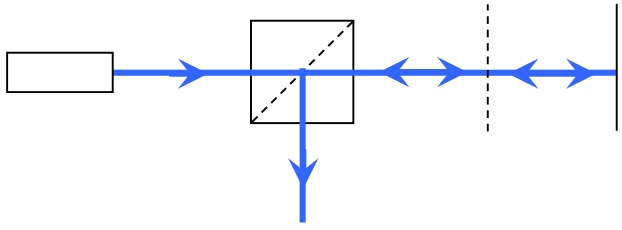
▪ Amplitude-splitting interferometers ▪



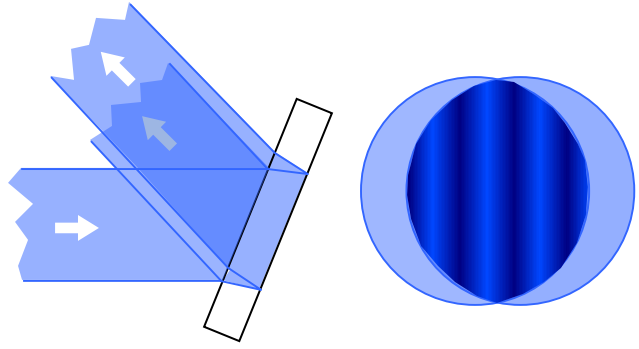
Michelson (Twyman-Green)



Mach-Zehnder

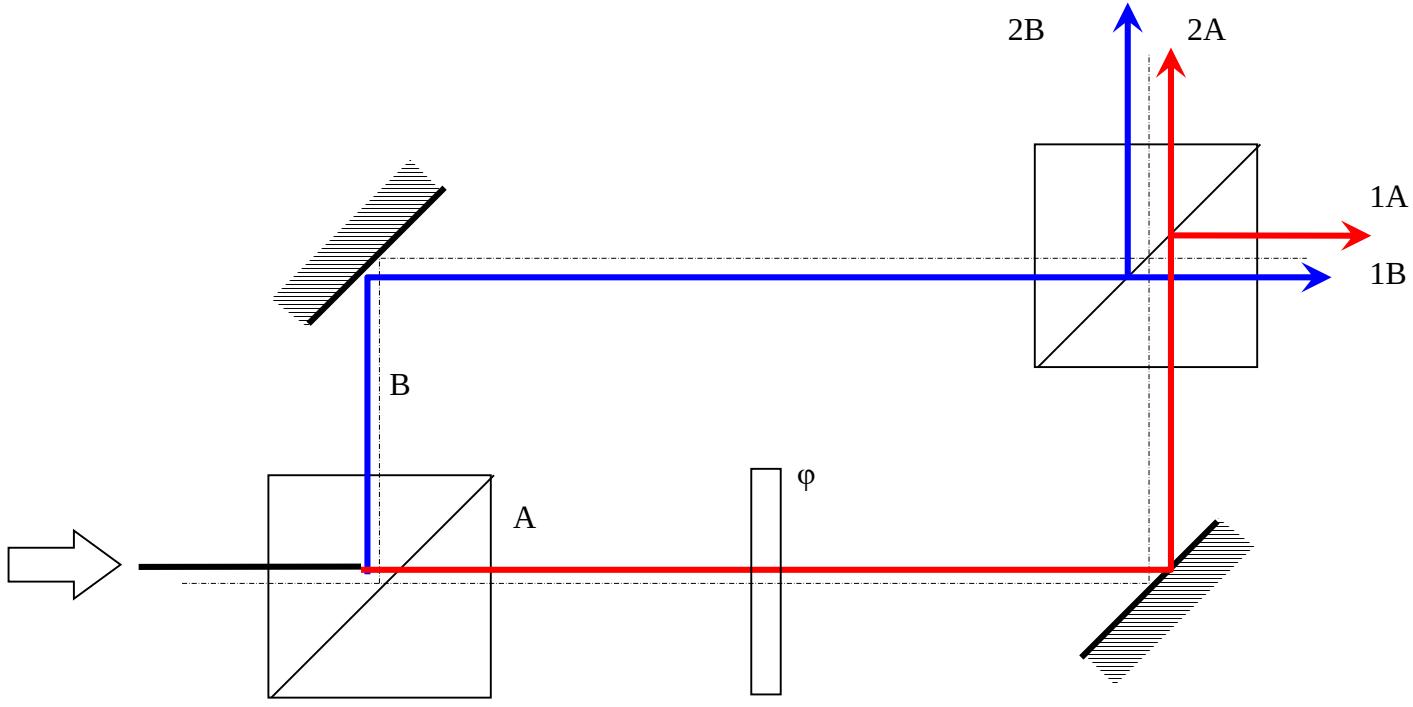


Fizeau



Shearing

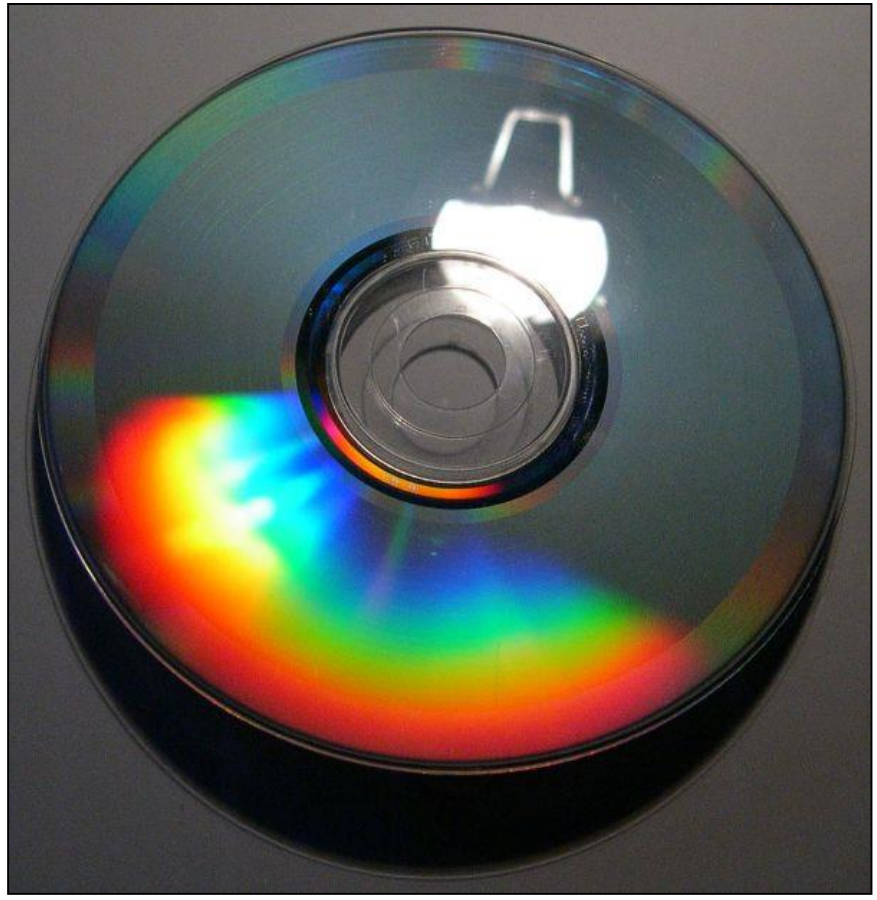
▪ What happens to light? ▪



▪ Interference examples (multiple-beam) ▪

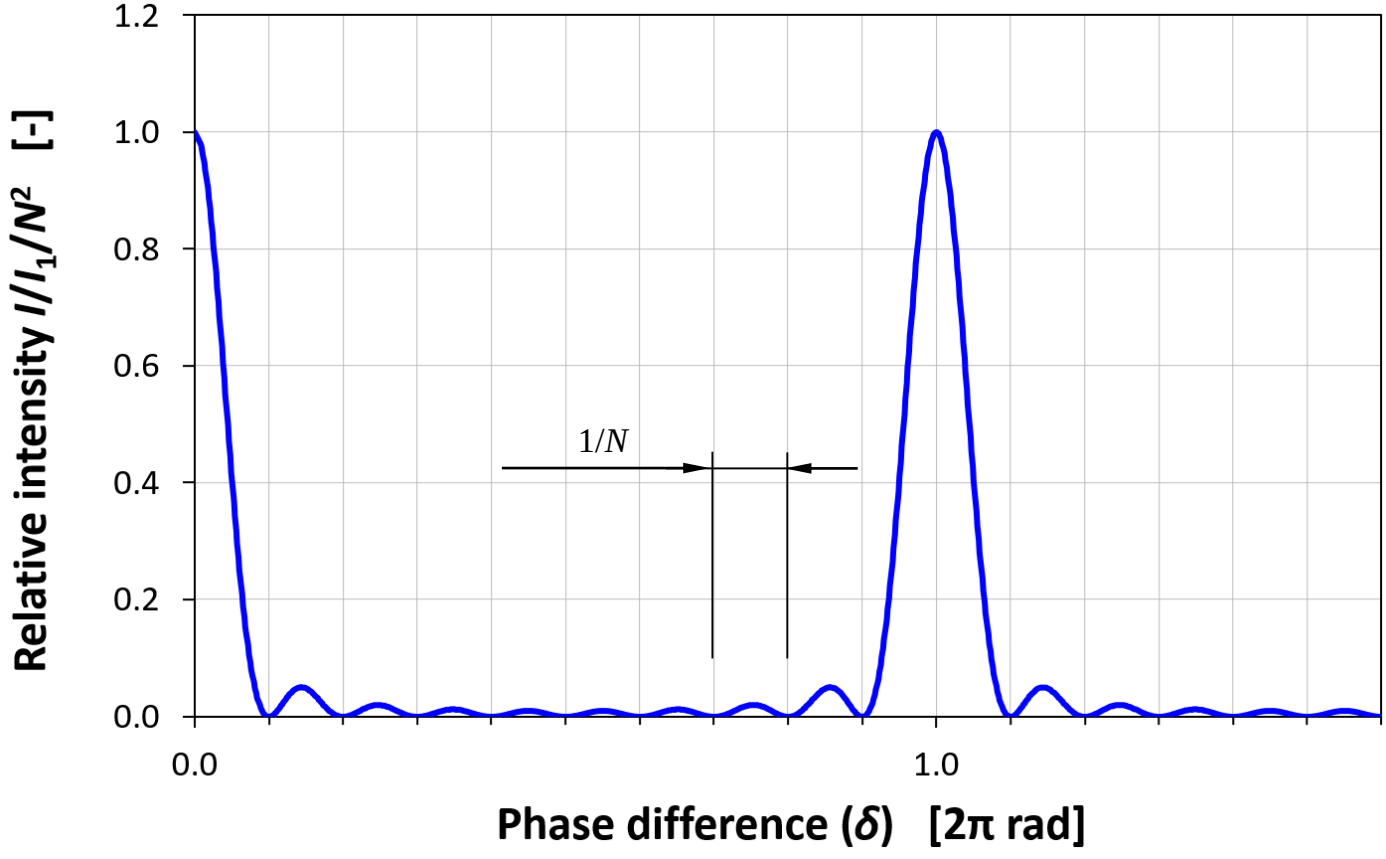


Antireflection coating (AR)

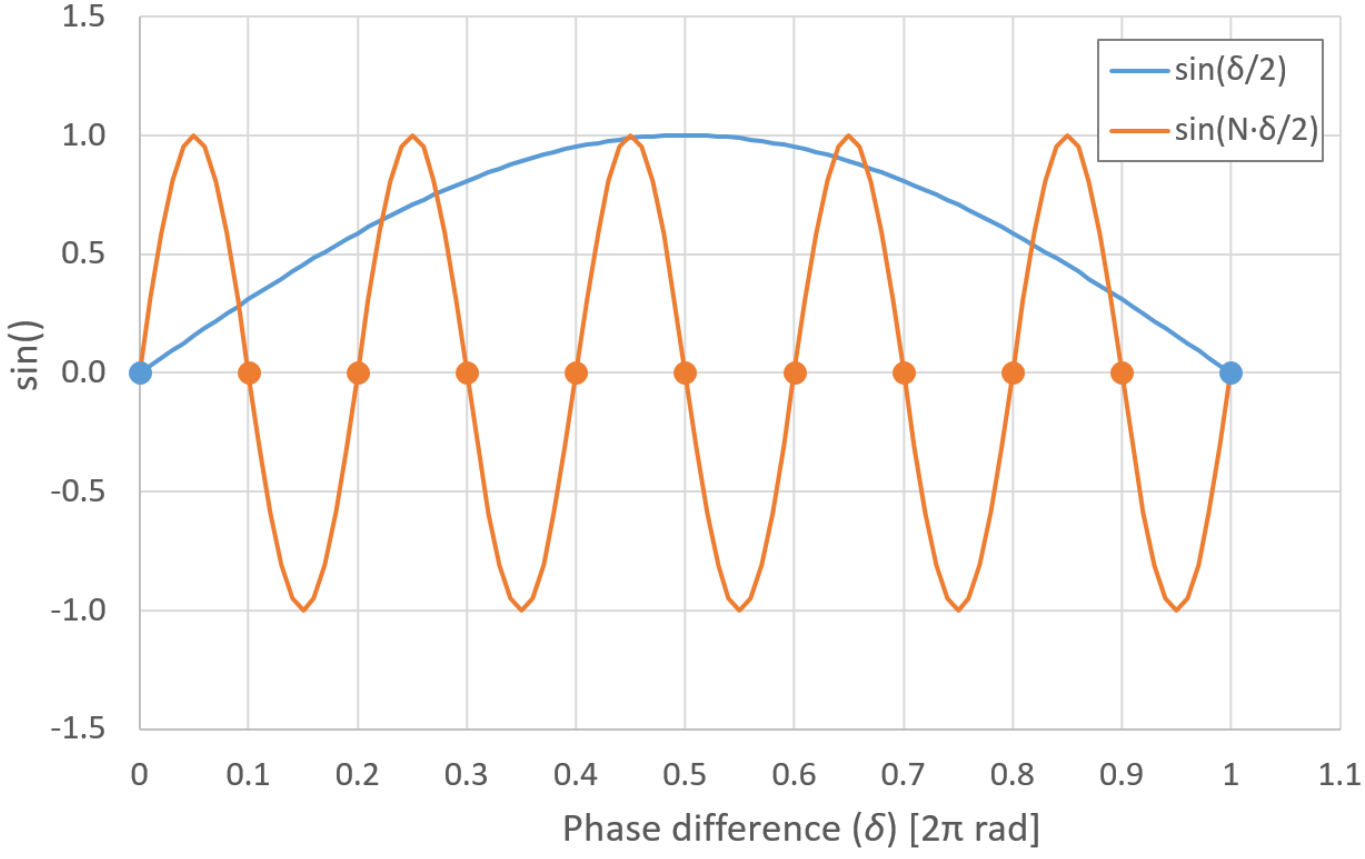


Diffraction grating (CD)

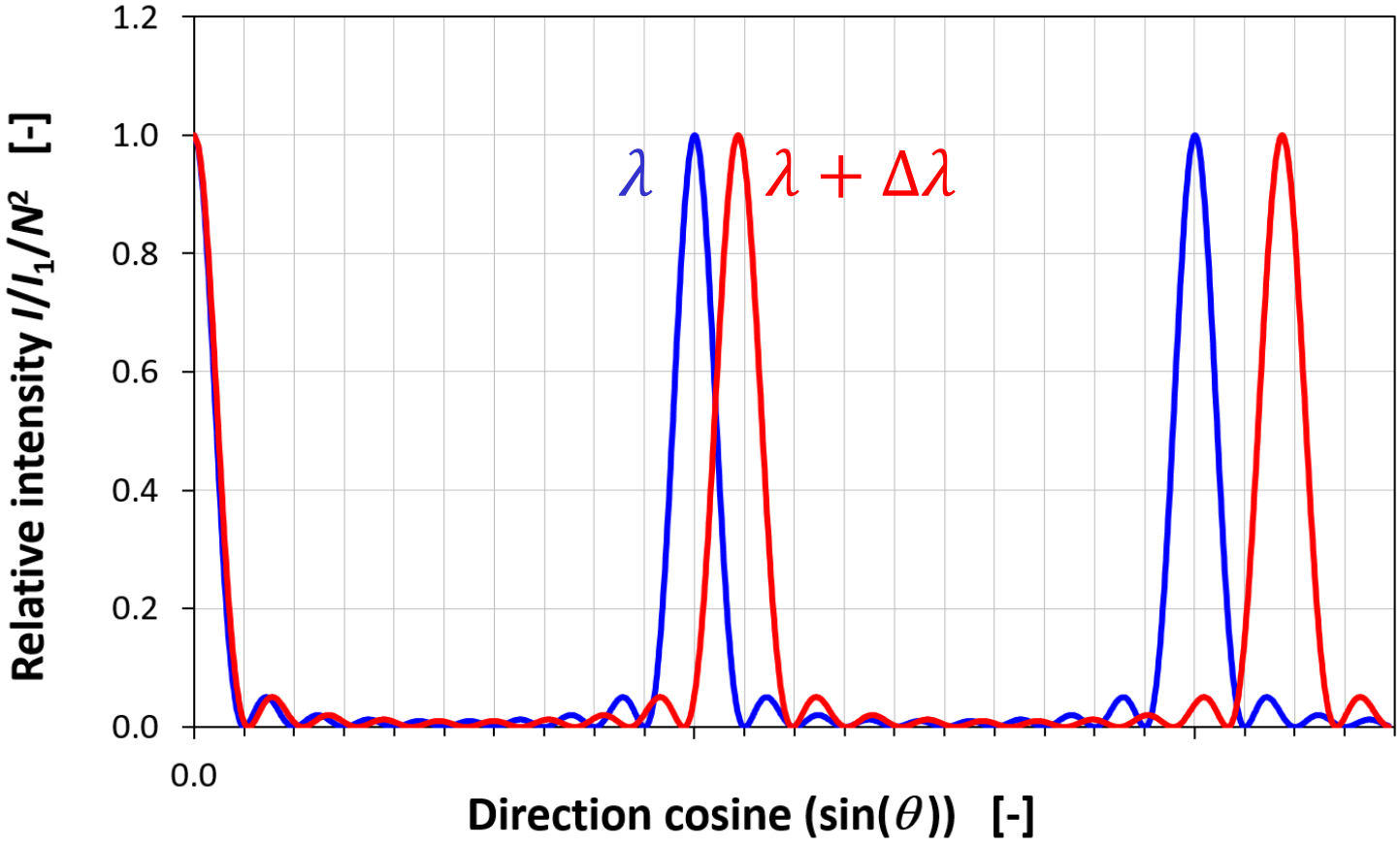
▪ Multiple-beam interference ▪



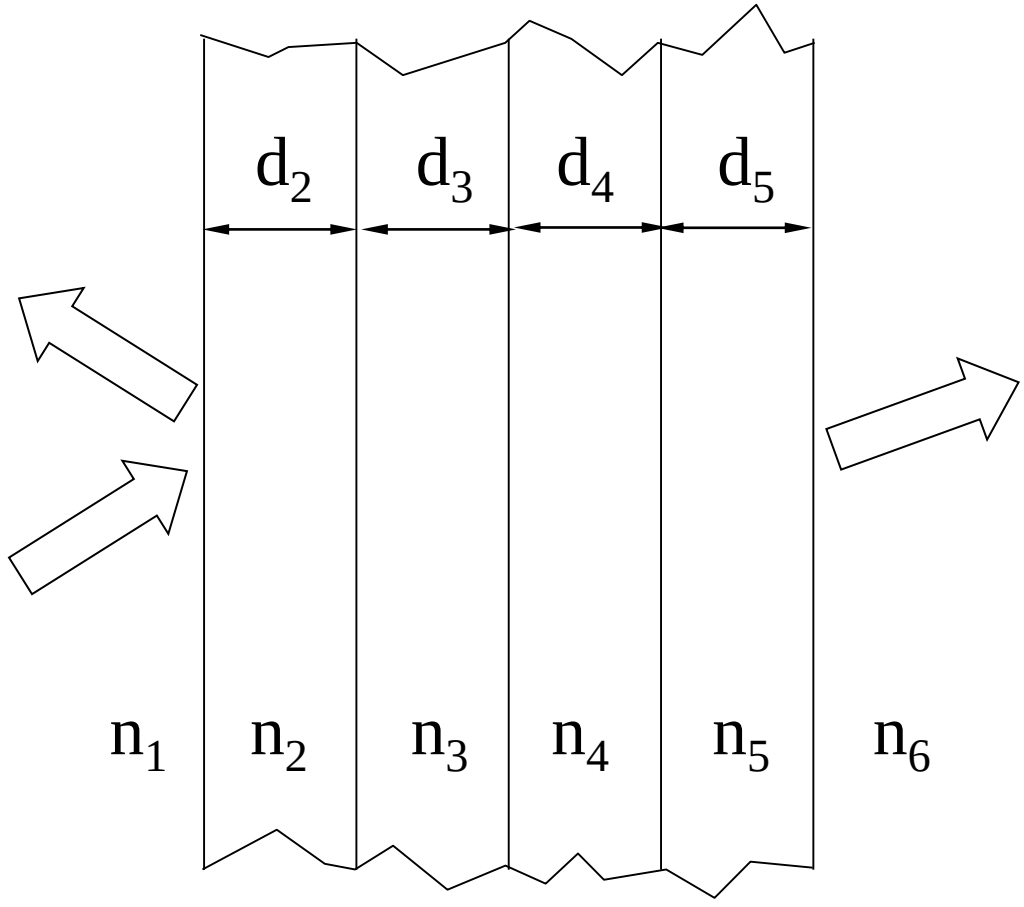
▪ Clarification on multiple-beam interferograms ▪



▪ Resolution of diffraction gratings ▪



▪ Matrix description of multilayers ▪



(e.g. Photonics subject)